Lab 4 - Configuring OSPF Authentication, NTP, Syslog and SSH

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
RT1	G0/0/1	172.16.1.1	255.255.255.0	N/A
	S0/1/0	10.1.1.1	255.255.255.252	N/A
RT2	S0/1/0 (DCE)	10.1.1.2	255.255.255.252	N/A
	S0/1/1 (DCE)	10.2.2.2	255.255.255.252	N/A
RT3	G0/0/1	172.16.3.1	255.255.255.0	N/A
	S0/1/1	10.2.2.1	255.255.255.252	N/A
Server1	NIC	172.16.1.200	255.255.255.0	172.16.1.1
Server2	NIC	172.16.1.201	255.255.255.0	172.16.1.1
Admin-PC	NIC	172.16.3.100	255.255.255.0	172.16.3.1

Objectives

- Configure OSPF SHA256 authentication.
- Configure NTP.
- Configure routers to log messages to the syslog server.
- Configure RT1 to support SSH connections.
- Lock down a router using AutoSecure and verify the configuration.

Background / Scenario

In this activity, you will configure OSPF SHA authentication for secure routing updates.

The NTP Server is the master NTP server in this activity. You will configure authentication on the NTP server and the routers. You will configure the routers to allow the software clock to be synchronized by NTP to the time server. Also, you will configure the routers to periodically update the hardware clock with the time learned from NTP.

The Syslog Server will provide message logging in this activity. You will configure the routers to identify the remote host (Syslog server) that will receive logging messages.

You will need to configure timestamp service for logging on the routers. Displaying the correct time and date in Syslog messages is vital when using Syslog to monitor a network.

You will configure RT1 to be managed securely using SSH instead of Telnet. The servers have been pre-configured for NTP and Syslog services respectively.

You will also use automated security features on router R3.

The routers have been pre-configured with the following passwords:

- Enable password: **ciscoenpa55**
- Password for vty lines: **ciscovtypa55**

Part 1: Configure OSPF SHA Authentication

Step 1: Test connectivity. All devices should be able to ping all other IP addresses.

Step 2: Create a SHA authentication key chain on RT1 for use with OSPF

a. Create a key chain called **OSPFSHA** on RT1 using the **key chain** command in order to enter key chain configuration mode.

b. Assign the key chain number **10** and set the password as **sha256OSPF** using the **key** and **key-string** commands respectively.

c. Specify SHA256 authentication with the **cryptographic-algorithm** command.

Step 3: Assign the authentication key to the OSPF interfaces on RT1

Assign the previously created authentication key to the serial interface on RT1 using the **ip ospf authentication key-chain** command.

QUESTION 1 Wait for approx. 30 seconds and then attempt to ping from the Admin-PC to either of the servers. Were the pings successful and, if not, why?

Before proceeding, go to the "Lab 4 - Configuring OSPF Authentication, NTP, Syslog and SSH - QUESTIONS" quiz on the Brightspace page and enter your answer for question 1. Leave the quiz open while you complete the rest of the lab sheet.

Step 4: Configure SHA authentication for OSPF on RT2 and RT3

Repeat steps 2 and 3 on RT2 and RT3 for all the active serial interfaces.

Step 5: Verify configurations.

a. Verify the SHA256 authentication configurations using the commands **show ip ospf interface**.

b. Verify end-to-end connectivity.

QUESTION 2 Can Admin-PC ping the servers now?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 2. Leave the quiz open while you complete the rest of the lab sheet.

Part 2: Configure NTP

Step 1: Enable NTP authentication on Server2.

a. On **Server2**, click **NTP** under the Services tab to verify NTP service is enabled.

b. To configure NTP authentication, click **Enable** under Authentication. Use key **5** and password **md5pa$$NTP** for authentication.

Step 2: Configure RT1, RT2, and RT3 as NTP clients.

a. Configure the three routers to obtain their time information from the NTP server.

b. Verify client configuration using the command **show ntp status**. (Note: You may have to wait several minutes to verify. You can click the "Fast Forward Time" button in the bottom corner to speed it up.)

QUESTION 3 Is the clock showing as synchronised?

QUESTION 4 What stratum number is the router currently showing as?

QUESTION 5 Based on this stratum number, what is the stratum number of the NTP server that the router is obtaining its time from?

Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 3-5. Leave the quiz open while you complete the rest of the lab sheet.

Step 3: Configure routers to update hardware clock.

Configure **RT1, RT2, and RT3** to periodically update the hardware clock with the time learned from NTP.

Exit global configuration and verify that the hardware clock was updated using the command **show clock**.

Step 4: Configure NTP authentication on the routers.

Configure NTP authentication on **RT1, RT2, and RT3** using key **5** and password **md5pa$$NTP**.

Part 3: Configure Routers to Log Messages to the Syslog Server

Step 1: Configure routers to timestamp log messages.

Configure timestamp service for logging on the routers using the **service timestamps** command. *Hint: Try using the context sensitive help ("?") to figure out how to complete this command as necessary.*

Step 2: Configure the routers to identify the remote host (Syslog Server) that will receive logging messages.

Configure the routers to send log messages to the Syslog server and exit back to enable mode. The router console will display a message that logging has started.

Step 3: Verify logging configuration.

Use the command **show logging** to verify logging has been enabled.

Step 4: Examine logs of the Syslog Server.

From the **Services** tab of the **Syslog Server**'s dialogue box, select the **Syslog** services button. Observe the logging messages received from the routers.

Disable the LAN interface on either RT1 or RT3 and verify that a log message is sent to the Syslog server (make sure to re-enable the interface again).

QUESTION 6 What is the severity level of the log message(s) displayed when the interface is manually disabled?

QUESTION 7 What is the MNEMONIC on the log message?

Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 6 & 7. Leave the quiz open while you complete the rest of the lab sheet.

Part 4: Configure RT1 to Support SSH Connections

Step 1: Configure a minimum password length.

Configure a minimum password length of **12** character on RT1.

Step 2: Configure a maximum number of failed login attempts before new attempts are disabled.

Configure RT1 to disable login attempts for a minute and a half when 5 failed login attempts are inputted within 30 seconds.

Step 3: Configure log messages to be sent for unsuccessful login attempts.

Configure RT1 to send log messages for unsuccessful login attempts.

Step 4: Configure a domain name.

Configure a domain name of **netsec.com** on **RT1**.

Step 5: Configure users for login to the SSH server on RT1.

Create a user ID of **AdminSSH** and a secret password of **ciscosshpa55**.

Step 6: Configure the incoming vty lines on RT1.

Use the local user accounts for mandatory login and validation. Accept only SSH connections.

Step 7: Erase existing key pairs on RT1.

Any existing RSA key pairs should be erased on the router.

Note: If no keys exist, you might receive this message: `% No Signature RSA Keys found in configuration.`

Step 8: Generate the RSA encryption key pair for RT1.

The router uses the RSA key pair for authentication and encryption of transmitted SSH data. Configure the RSA keys with a modulus of **2048**.

Step 9: Verify the SSH configuration.

Use the **show ip ssh** command to see the current settings.

QUESTION 8 What is the current SSH version?

QUESTION 9 What is the currently timeout (in seconds)?

QUESTION 10 What is the current number of authentication retries?

Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 8-10. Leave the quiz open while you complete the rest of the lab sheet.

Step 10: Configure SSH timeouts and authentication parameters.

The default SSH timeouts and authentication parameters can be altered to be more restrictive. Set the timeout to **100** seconds, the number of authentication retries to **1**, and the version to **2**.

Issue the **show ip ssh** command again to confirm that the values have been changed.

Step 11: Attempt to connect to RT1 via Telnet from Admin-PC.

Open the Desktop of **Admin-PC**. Select the Command Prompt icon. From **Admin-PC**, enter the command to connect to **RT1** via Telnet.

```
PC> telnet 10.1.1.1
```

Step 12: Connect to RT1 using SSH on Admin-PC.

Open the Desktop of **Admin-PC**. Select the Command Prompt icon. From **Admin-PC**, enter the command to connect to RT1 via SSH. When prompted for the password, enter the password configured for the administrator **ciscosshpa55**.

```
PC> ssh -l AdminSSH 10.1.1.1
```

QUESTION 11 Were the Telnet or SSH successful?

Step 13: Verify failed login attempts are being monitored.

a. Close the SSH connection to RT1 on **Admin-PC** (if still open) and attempt to make a new connection but this time deliberately misspell the password.

b. Open the **Services** tab of the **Syslog Server**'s dialogue box, select the **Syslog** services button and verify if a log message has been received from RT1 relating to the failed login attempt.

c. Use the command **show login failures** on **RT1** to display information about failed login attempts.

QUESTION 12 What is the *source IP address* showing on the output on RT1 for the most recent failed login attempt?

Before proceeding, return to the quiz on the Brightspace page and enter your answers for questions 11 & 12. Leave the quiz open while you complete the rest of the lab sheet.

Part 5: Configure Automated Security Features

In this part, you will use AutoSecure to secure RT3. By using a single command in CLI mode, the AutoSecure feature allows you to disable common IP services that can be exploited for network attacks. It can also enable IP services and features that can aid in the defense of a network when under attack. AutoSecure simplifies the security configuration of a router and hardens the router configuration.

Step 1: Use the AutoSecure Cisco IOS feature on RT3.

a. Enter privileged EXEC mode using the **enable** command.

b. Issue the **auto secure** command on RT3 to lock down the router. The connection to RT2 represents a WAN connection, so assume that RT3 S0/1/1 is connected to the internet when prompted by the AutoSecure questions. Respond to the AutoSecure questions as shown in the following output. The responses are bolded.

```
RT3# auto secure
--- AutoSecure Configuration ---
*** AutoSecure configuration enhances the security of
the router, but it will not make it absolutely resistant
to all security attacks ***

AutoSecure will modify the configuration of your device.
All configuration changes will be shown. For a detailed
explanation of how the configuration changes enhance security
and any possible side effects, please refer to Cisco.com for
Autosecure documentation.
At any prompt you may enter '?' for help.
Use ctrl-c to abort this session at any prompt.

Gathering information about the router for AutoSecure

Is this router connected to internet? [no]: yes
Enter the number of interfaces facing internet [1]: [Enter]
Interface  IP-Address  OK? Method Status  Protocol
GigabitEthernet0/0/0  unassigned  YES NVRAM  administratively down down
```

```
GigabitEthernet0/0/1  172.16.3.1  YES manual up  up
Serial0/1/0  unassigned  YES NVRAM  administratively down down
Serial0/1/1   10.2.2.1  YES manual up  up
Vlan1  unassigned  YES unset  administratively down down

Enter the interface name that is facing internet: Serial0/1/1

Securing Management plane services...

Disabling service finger
Disabling service pad
Disabling udp & tcp small servers
Enabling service password encryption
Enabling service tcp-keepalives-in
Enabling service tcp-keepalives-out
Disabling the cdp protocol

Disabling the bootp server
Disabling the http server
Disabling the finger service
Disabling source routing
Disabling gratuitous arp

Here is a sample Security Banner to be shown
at every access to device. Modify it to suit your
enterprise requirements.

Authorized Access only
  This system is the property of So-&-So-Enterprise.
  UNAUTHORIZED ACCESS TO THIS DEVICE IS PROHIBITED.
  You must have explicit permission to access this
  device. All activities performed on this device
  are logged. Any violations of access policy will result
  in disciplinary action.


Enter the security banner {Put the banner between
k and k, where k is any character}:
# This service is restricted to authorised users only. All activities on this system are logged. #
Enable secret is either not configured or
 is the same as the enable password
Enter the new enable secret: ciscoenpa55
Confirm the enable secret: ciscoenpa55
Enter the new enable password: 55ciscoenpa
Confirm the enable password: 55ciscoenpa

Configuration of local user database
Enter the username: AdminSSH
Enter the password: ciscosshpa55
Confirm the password: ciscosshpa55
Configuring AAA local authentication
Configuring console, Aux and vty lines for
local authentication, exec-timeout, transport
Securing device against Login Attacks
Configure the following parameters
```

```
Blocking Period when Login Attack detected: 90

Maximum Login failures with the device: 5

Maximum time period for crossing the failed login attempts: 30

Configure SSH server? [yes]: [Enter]

Enter the host name: RT3
Enter the domain-name: netsec.com
Disabling mop on Ethernet interfaces

Securing Forwarding plane services...

Enabling CEF (This might impact the memory requirements for your platform)
Enabling unicast rpf on all interfaces connected
to internet

Configure CBAC Firewall feature? [yes/no]: no
Tcp intercept feature is used prevent tcp syn attack
on the servers in the network. Create autosec_tcp_intercept_list
to form the list of servers to which the tcp traffic is to
be observed
Enable tcp intercept feature? [yes/no]: no

This is the configuration generated:

!
service password-encryption
no cdp run
access-list 100 permit udp any any eq bootpc
banner motd # This service is restricted to authorised users only. All activities on this
system are logged. #
enable secret 5 $1$mERr$TfFTxE.mmb5O5BVC56ndL0
enable password 7 0874194D000A061817051C05
username AdminSSH password 7 0822455D0A1616041A1B0D517F
aaa new-model
aaa authentication login local_auth local
line con 0
 login authentication local_auth
 exec-timeout 5 0
 transport output telnet
line vty 0 4
 login authentication local_auth
 transport input telnet
service timestamps debug datetime msec
service timestamps log datetime msec
logging trap debugging
logging console
logging buffered
line vty 0 4
 transport input ssh
 transport input telnet
hostname RT3
ip domain-name netsec.com
```

```
ip access-list extended 100
 permit udp any any eq bootpc

Apply this configuration to running-config? [yes]: [Enter]

Applying the config generated to running-config
The name for the keys will be: test.test


% The key modulus size is 1024 bits
% Generating 1024 bit RSA keys, keys will be non-exportable...
*Mar  1 22:56:41.001: %SYS-3-CPUHOG: Task is running for (2007)msecs, more than
(2000)msecs (0/0),process = crypto sw pk proc.
-Traceback= 0x824198E0 0x82419FC4 0x8283C238 0x82866AD8 0x828667A8 0x82865D34 0x
828660F4 0x82866510 0x802335D4 0x80236D80 [OK]
RT3#
```

Note: On a productions system, the questions asked would vary depending on the features on the IOS image and device.

Step 2: Correct the OSPF authentication.

The AutoSecure configuration will have caused the OSPF authentication password to have been altered. In order to fix this, please re-do Part 1 Step 2.

If you have correctly configured all parts of the lab your activity score should now be showing as 100%. If so, click on "check results" in the activity window. Return to the Brightspace quiz one last time and enter the code into the appropriate question box (Q13) of the quiz.

You have completed the lab - please submit the Brightspace quiz.